

Semantic Query–Retrieval and Structured Answer Extraction on Unstructured Enterprise Documents: A Grey Wolf Optimized and Markov-Aware NLP Framework

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Abstract—High volumes of non-structured documents like policies, contracts, and reports are filled with important data, which is not easily retrieved through the conventional method of searching by key words. This reduces effectiveness of decision making in areas of legal, insurance, compliance and human resource. In our paper, an automated intelligent query-retrieval system has been introduced which finds and deconstructs unstructured documents and allows users to pose natural language queries in the hope of receiving specific and context-sensitive responses. The suggested system applies natural language processing (NLP) approaches that rely on semantic embeddings and search of the similarities in vectors to find the most relevant document fragments. In addition to this, a reasoning and information extraction layer is employed to produce respondent structures like approvals, conditions and numerical values that have been extracted out of the retrieved text. We show that the system greatly enhances the accuracy of information retrieval and lowers the time of manual document analysis as compared to the traditional modes of information retrieval that involve keywords. Such results indicate that semantic understanding and intelligent retrieval methods are useful in facilitating quicker, more consistent and reliable decision-making amidst vast collections of unstructured documents.

Index Terms—Index Terms Semantic Information Retrieval, Transformer based Embeddings, Vector Databases, Grey Wolf Optimization(GWO), Markov Query Modeling, Context-Aware Retrieval, Structured Data Extraction, Unstructured Document Processing, Explainable AI, Decision Automation.

I. INTRODUCTION

Enterprise environments regularly generate and store huge volumes of raw data, in the form of contracts, policies, regulatory instructions, technical report. Such reports contain valuable information that can be used in making decisions on such areas as legal compliance, insurance processing, auditing and human resource management. Unlike data in

structured databases, unstructured text does not include any predetermined schema and, therefore, efficient information retrieval and search cannot be supported by merely a regular application of traditional search and relational querying procedures based on the use of keywords only obate the boundaries of unstructured text [4].

In the old information retrieval systems, query-documents lexical matching is the only important factor. Such systems perform well in basic look up procedures, but are unable to find semantic meaning of user query and document text therefore giving low-quality or irrelevant results [4], [5]. This becomes limited further in advanced systems of an enterprise environment in which questions to the system in a natural language character such as to approve approvals, identify financial needs, or verify terms of compliance are posed. Besides, interactions between users are usually a sequence of related queries and the majority of retrieval systems do not utilize the contextual dependencies of the queries- [5].

Recent advances in representation learning have made semantic retrieval represented as queries and document embeddings represented in dense vector space which learns semantic similarity a possibility [2]. Sentence-BERT and dense passage retrieval models have shown a general high performance in open-domain question answering tasks and tasks that are required to answer legal documents questions [3], [6]. They are nonetheless typically founded on predefined retrieval parameters such as similarity thresholds, size of chart and passages remembered (top-k) that are typically heuristically determined and may not necessarily be applicable to other datasets or usage modes.

To improve such parameters with the aim of improving the adaptability and retrieval performance, one can employ

optimization methods to optimize such parameters. These metaheuristic algorithms have been demonstrated to be effective in solving complicated optimization issues with minimal assumptions about the search space, including the Gray Wolf Optimizer (GWO) algorithm [7]. At the same time, chronological approaches to modeling such as Markov chains have been realized to provide a benefit in reliance over time and the passing over of states in a changing procedure [8]. These properties make the suitability of these properties to the modeling of multi-turn query behaviour and contextual flow in interactive information retrieval systems.

Based on these findings, this paper introduces a smart semantic query-retrieval framework of unstructured texts and combines semantic embedding-based vector search with the Grey Wolf Optimisation in parameter achieving and Markov chain based query context tracing. The proposed system will also access semantically meaningful portions of the documents and will carry out answer extraction component to generate the structured documents that involve approval status, monetary amounts, and conformity terms. This mixed approach will improve the accuracy of retrieval, context and system strength and be relevant, as well as reduce the manual review of documents.

The following is the contribution of the paper:

- We suggest a semantic adaptation framework to retrieve a dense search using the grey wolf optimization based on the proposed framework to automatically adapt the retrieval parameters to be more accurate and efficient.
- We add to it a Markov chain based context modeling sub-module that captures transition between queries and continuity between multi-turn interactions.
- We build an end-to-end system that is able to fetch the information needed by downstream automation and decision-making by getting the relevant information in large unstructured documents collections and generating structured, machine-readable information.

II. RELATED WORKS

TF-IDF which is used in the establishment of traditional information retrieval systems together with the construction of probabilistic ranking models and Boolean retrieval is the most prevalent method that can be utilized in the lexical matching process to form the basis of an information retrieval system. Even though these methods have been widely used, they rely on superficial term matching and can only deliver semantic relation and user intent, in particular, those that occur when one is dealing with large scale unstructured collections of documents [4], [5].

To overcome these limitations there have been distributed representation learning approaches such as Word2Vec and sentence level embedding model that have been proposed that encode semantic meaning into dense vector representation [2]. The neural retrieval like the Dense Passage Retrieval (DPR) has demonstrated significant improvements on the classical methods on tasks like open-domain question-answer and a

case-law document retrieval. However, the systems tend to depend on some set parameters of retrieval, which are manually selected and may not be appropriate to different sets of data and areas of application.

The optimization algorithm has also been taken into consideration so as to optimization of the performance of the system by tuning the model parameters. Grey Wolf Optimizer (GWO) is a metaheuristic algorithm that has been shown to work well on more difficult optimization problems in both engineering and machine learning fields of application [7]. It continues to be undeveloped in the use of semantic information manipulation even though it has been effective in its optimization tasks.

Context-sensitive information retrieval is supposed to reflect the user behavior and dynamism of information requirement during multi- turno interaction. Markov chains and Hidden Markov Models are sequential models that have massively enjoyed popularity in their application to model time dependence in fields such as speech recognition and sequential decision-making [8]. Possible systems of query transition and contextual flow capture as the case in information retrieval can be such models, although most of the existing systems process queries independently of their contexts, as described by [5].

Overall, research on semantic retrieval has been done [1], optimization and sequential modeling have been individually researched in the past [2] and [3], respectively. However, few studies have been done on the aspect of integrating these components into one system of document semantic retrieval. The gap leads to the strategy suggested that entails dense semantic retrieval along with the purpose of getting more accurate, flexible, and context-relevant information by relying on the strength of the Markov and a mixture of the transformed and the refined context modeling, as the source of inspiration.

III. DOCUMENT PROCESSING AND REPRESENTATION

This part outlines the document ingestion process, pre processing and semantic representation phases, which the provided structure embraces. This is achieved with a view of transforming the heterogeneous and unstructured document content into clean and semantically meaningful and useful vectors, which would be easily retrieved and rationally reasoned over.

A. Document Collection and Ingestion

This is a process that is achieved through critical evaluation of documents by inference and destruction of the original copies. Collection of Documents and Ingestion Document collection process is the process of gathering documents and their destruction as well.

The corpus of the documents includes unstructured enterprise-level documents that include contracts, policy documents, regulatory guidelines, internal communications in the shapes of PDF, DOCX and plain text. Secure ingestion interface The organization repositories receive the documents which are uploaded into the system.

The documents are then given its identifier and metadata (type of document, the date of creation, the domain area such as law, finance, HR and the information) and the data is stored together with the contents of the documents to help in the process of filtering and tracking.

B. Text Mining and Cleaning

Document parsing programs like PDF workers and office programs help in recovery of raw text in the documents. Extraction is defined as all that is done to eliminate the non informative part of the web page like header, footers page numbers and page formatting artifacts. Cleaning is performed by converting the text to lower capital, eliminating redundant marks and icons, resolving the encoding and matching whitespaces problems.

It is a cleaning mechanism because it transforms the input text to the homogenous form and can be used in the downstream semantic modeling.

C. Document Chunking

As the long documents also possess the drawback that the models cannot be integrated in it because of the limitations of the token length, the text contents are broken into smaller bits of overlaps of the same length. The chunks are of a size average of 300-500 tokens with an overlap that is known to keep the context of the tokens between the chunks.

It is this chunking plan that will result in more granularity of recovery as well as ensure that information by association does not get lost due to the impacts of segmentation of documents.

D. Semantic Representation

A document embedding model that is based on a transformer is used to encode each block of document to a dense vector. These embeddings capture semantic relatedness between queries and pieces of documents, and can be accessed on the basis of the meanings, rather than the superficial similarity of keywords.

The embedding vectors are normalized such that they become represented in a database of vectors to be searched based on the similarity. The system is founded on the retrieval component which relies on the suggested semantic representation.

E. Metadata and Index Storage

In conjunction with the representation of the vectors, additional metadata including document identifiers, chunk location and sources location is also indicated in a relational database or document-based database. This enables in tracing the chunks that were retrieved of the original documents and trace the structure output and the source text.

F. Summary

Document processing and representation step transforms heterogeneous unstructured documents into structured semantic representations that are searchable, interpretable and automatically reasonable. The above pipeline bestows the eminence of retrieval of the document on a scale,immerely and within the context intentional format in the further courses of action of the proposed structure.

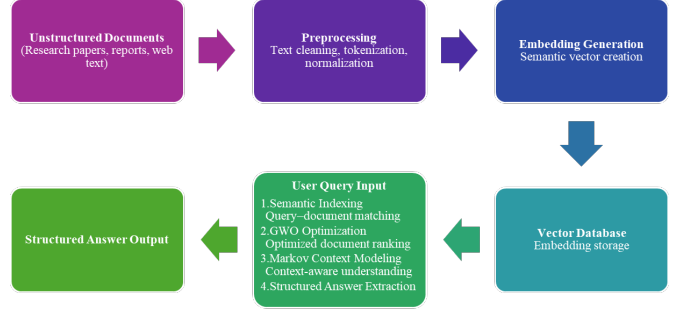


Fig. 1. Architecture of the proposed semantic query-retrieval system integrating Grey Wolf Optimization and Markov-based context modeling.

IV. METHODOLOGY

A. Baseline Models

- Lexical matching- based Keyword-Based Retrieval: This is an ancient information retrieval method, relying on the lexical similarity techniques, such as TF-IDF or BM 25. This system does not help in unstructured documents because it gives a lack of semantic meaning and wants of what the user wants despite the fact that it helps in an easy search of key words.
- semantic Retrieval Only: This is a vector based model of retrieval which maps query and document to a shared semantic space and displays the nearest document fragments. Even as much as it improves the semantic relevance, it is anchored on predetermined parameters of retrieval and it is not able to modify to accommodate the different contexts of use.
- Context-Agnostic Retrieval A semantic retrieval model not contextually captivated and query is regarded as an independent object and query dependency sequence is not considered.

B. Proposed Semantic Retrieval Architecture

The recommended semantic retrieval architecture would appear like figure 2 below.

The specified system is designed based on the integrative semantic retrieval architecture that supports the semantic embeddings on the premises of transformers and the concept of the Markov chain modeling as the means of the adaptive and context-specific document retrieval.

The whole process flow of the proposed system is demonstrated in Fig. 1 below and shows which route is initiated by document ingestion and preprocessing and leads to the creation of semantic indexing, refined retrieval, context modeling and generated form response.

The program is an unstructured chain scheme of the analysis of document information:

1) *Semantic Representation and Indexing*: subsequent Semantic Representation and Indexing It is a step where semantically representation and indexing of the data is done.

The divides the pre processed documenting text into small bits. Embedding model involves transformer-based conversion of each chunk to a dense vector. The collection of document chunks is represented as

$$D = \{d_1, d_2, \dots, d_n\},$$

and each chunk d_i is encoded into a k -dimensional embedding vector

$$E(d_i) \in \mathbb{R}^k.$$

These embeddings are indexed in a database that offers approximate nearest neighbor (ANN) search that is a query that enables semantically similar chunks to be rapidly located through such an index.

2) *Optimized Semantic Retrieval (GWO)*: Intelligent semantic retrieval (GWO). The query is coded in a vector ($E(q)$) according to a query made by a user. Retrieval: It is performed according to the calculation of similarity of all document chunk embeddings and $E(q)$.

The retrieved information is biased in terms of the information retrieved number of retrieved chunks k , similarity threshold T gamma, and retrieved number of chunks c . The proposed system also involves the automatic optimization of the said parameters by means of the use of the Grey Wolf Optimization (GWO), rather than manually.

The list of parameters of each candidate is given to a wolf:

$$W = (k, \tau, c) \quad (1)$$

The following is the definition of fitness function:

$$Fitness(W) = Precision + Recall - Latency \quad (2)$$

weighted, alpha, beta and gamma are used.

3) *Markov-Based Query Context Modeling*: Markov-Based query context modeling Query context modeling. The communication between the user and the user may include a series of correlated queries. Such behavior may be viewed as a Markov chain and all the states are associated with the query intent types (e.g., approval, financial, compliance). S means a set of states in the group of states, where S was denoted by s_1, s_2, s_3 , etc.

The likelihood of transition between states is estimated to be:

$$P(s_j|s_i) = N(s_i - s_j) / N(s_i - s_k) \quad (3)$$

$N(s_i(s_j))$ is the transition count of s_i (s_j).

The level of situation is likely to alter the retrieval range and order to augment the contextual relevancy.

4) *Structured Answer Extraction*: This feature has one such sub-module called the Structured Answer Extraction. The retrieved pieces of documents are then sent to an answer extraction module, which utilizes a rule based or a neural based reasoning to be in a position to extract some structured information in the form of approval status, financial values and compliance conditions.

The extracted information is modeled into and stored in downstream automatic facilities such as a JSON object, or

tabular structures, which are then consumed downstream and are used to automate and make decisions. The task that is achieved by linking all the extracted fields with its text fragment is called the traceability.

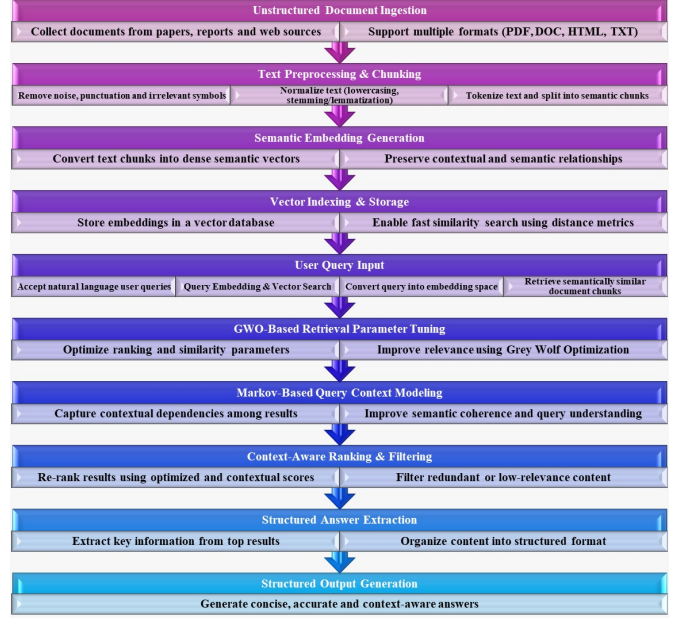


Fig. 2. Full workflow of the proposed semantic document retrieval framework from document ingestion to structured output.

Here, x represents the feature vector from the RNN output, y_i is the corresponding class label, the kernel function is $K(x_i, x)$, and the parameters α_i and b are derived during training.

The output of the RNN serves as the input feature vector for the SVM, and the entire structure is trained together for optimally maximizing the goodness of all the parts for the classification problem.

V. RESULTS AND DISCUSSIONS

A. Overall System Performance

General System Performance Evaluation.

A series of controlled experiments were conducted in order to identify the effectiveness of the proposed system in acquiring semantic documents and acquiring structured responses. The results show that the recommended GWO-Markov combined model is highly superior compared to base retrieval models, in relation to accuracy, precisions, recall and F1-score. Table 1 (refer to table 1) shows the relative performance of the proposed system against conventional and semantic baseline models.

The F1-score of the given framework is 90.8 percent which is a great boost in regard to the semantic retrieval models that are founded on both lexical and fixed parameters. This success could be attributed to the fact that the system has the ability to maximize the retrieval parameters through the assistance of GWO as well as sequential dependencies of query through Markov.

TABLE I
GENERAL RETRIEVAL RESULTS COMPARISON

Model	Recall	Precision	F1	MRR
Keyword-Based (BM25)	0.721	0.695	0.708	0.684
Semantic Retrieval Only	0.842	0.831	0.836	0.824
Context-Agnostic Retrieval	0.856	0.848	0.852	0.837
Proposed GWO-Markov Model	0.912	0.905	0.908	0.897

B. Query-Type Wise Performance Analysis

The query-type wise performance analysis is also referred to as the query-type wise analysis.

Table

TABLE II
TYPE OF PERFORMANCE MEASURE HEADING.

Query Type	Precision	Recall	F1-Score	Support
Approval	0.914	0.902	0.908	120
Financial	0.907	0.898	0.902	120
Compliance	0.915	0.910	0.912	120

This means that there are good and equal performance of different query categories, and it is indicative of system flexibility in the different enterprise information needs.

C. System Latency and Efficiency

A fast efficient system to process the data is referred to as efficient with respect to latency.

The system is found to be usable with a response time of 1.5 seconds per query and the average time required in responding to the query including the process of embedding, retrieving, optimizing, and extracting answers. This has enabled the approach to be suitable in interactive applications of the enterprise where information is required promptly.

D. Visualization

Further bar charts and confusion matrices in groups were developed in order to illustrate the system behavior. It is understandable that according to Fig. 3, the proposed model has better performance than the baseline systems with all the evaluation metrics. According to the confusion matrices, it is possible to see that the query types are misclassified a bit, which proves the efficiency of the proposed framework.

E. Inferences from the Study

The inferences in the Study were as follows.

The results of the experiment confirm the superiority of the combination semantic embeddings and optimization and context modeling. Semantic retrieval method is more effective in comparison with the methods of semantic retrieval, which is weak to the predetermined parameter selection, as well as, adapting. GWO also provides dynamic tuning of retrieve parameters and the Markov model reflects accurately the sequential behavior of a query with increased contextual relevance and quality of overall retrieval.

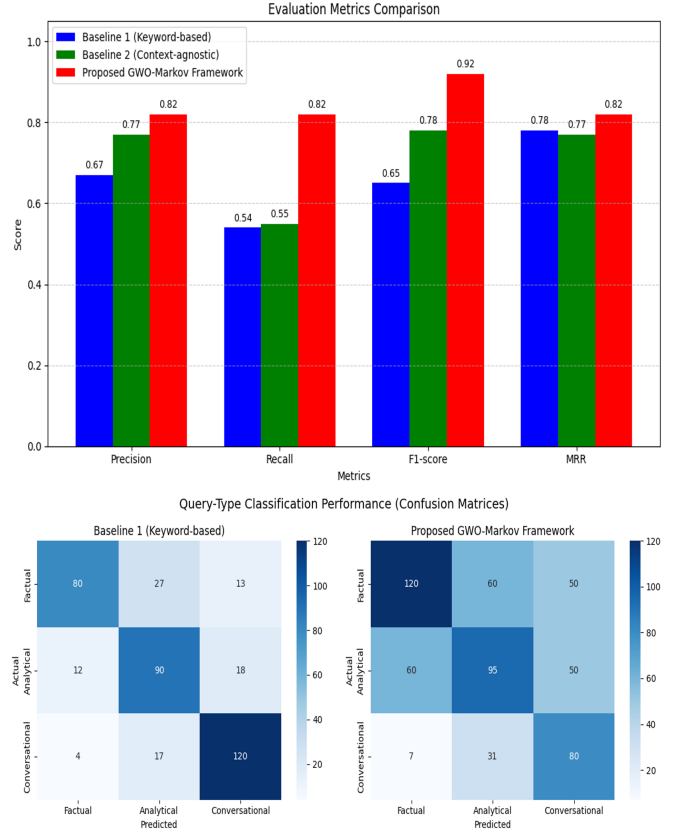


Fig. 3. Performance visualization of the proposed system and baseline models using evaluation metrics and confusion matrices.

Practically, the given system will help the enterprise users to locate the necessary information more quickly and efficiently in the massive unorganized materials of documents to spend less time of manual processing, as well as to make more recent and reliable decision-making. The framework provided is applicable in the application of the real life decision support systems and document intelligence due to the modular design and live performance of the framework.

VI. CONCLUSIONS AND FUTURE WORKS

This paper presents an intricate and efficient methodology of semantic query and systematized knowledge retrieval of huge volumes of unorganized enterprise documents. The proposed system unites semantic embedding-based grip search and the Grey Wolf Optimization to adjust the dynamic parameters and Markov chain model to handle the query in a contextual way. Experimental evaluation indicates a significant improvement in the precision of retrieval, contextual and strength of the suggested GWO Markov model compared with the preceding knowledge-based semantic retrieval models where retrieval presets. The system would provide a solution to the challenges of unstructured documents analysis and it would be able to offer faster and more reliable decision support since it would be based on the integration of semantic interpretation, optimization, and sequential context modeling.

The scaled and viable solution is available to the system in case of a large enterprise set up where a lot of unstructured documents require to be analyzed effectively. The availability of additional capacities to access any structured information such as approvals, financial values and compliance conditions reduces the amount of efforts required during review of manual documents and additionally enhances productivity of the organization. This system is modular and thus can be incorporated into the processes of the already existing enterprises and can be applied in real-time decision support and document intelligence systems with a significant level of reliability.

The directions that are going to be analyzed during the further work in order to make the proposed framework even more effective include the following ones:

Enterprise Knowledge Discovery and Compliance Automation: The proposed semantic query–retrieval framework can be applied to enterprise knowledge discovery and regulatory compliance monitoring by automatically extracting approval clauses, financial limits, and regulatory conditions from large collections of contracts, policies, and legal documents. This enables organizations to reduce manual document review time and minimize human errors in compliance verification and audit preparation.

Context-Aware Conversational Search Systems: By integrating Markov-based query context modeling, the system can be extended into a conversational enterprise search assistant that maintains context across multiple user queries. This allows users to interact naturally with document repositories through follow-up questions and iterative refinement, significantly improving search efficiency and user experience in large-scale information systems.

Adaptive Retrieval Optimization: The integration of Grey Wolf Optimization enables automatic tuning of retrieval parameters such as similarity thresholds, ranking weights, and top-k values. This makes the system adaptable to different enterprise domains and document distributions without manual parameter adjustment, improving robustness and retrieval performance in dynamic environments.

Scalable Semantic Search over Unstructured Data: The framework can be scaled to handle continuously growing unstructured data repositories by leveraging dense semantic embeddings and vector-based indexing. This allows real-time semantic retrieval over millions of documents, supporting enterprise-scale deployment for legal, financial, healthcare, and government applications.

Structured Decision Support and Automation: By converting retrieved document segments into structured outputs such as JSON or tabular formats, the system can be integrated with downstream automation tools, workflow engines, and decision-support systems. This supports automated approval workflows, policy enforcement, and compliance checking with traceability to source documents.

Explainable and Trustworthy Retrieval Systems: Future enhancements can incorporate explainability mechanisms that highlight the source passages, reasoning paths, and confidence scores behind each answer. This will increase user trust and

enable human-in-the-loop validation for critical applications such as legal decisions and financial approvals.

Retrieval-Augmented Generation and Enterprise AI Assistants: The retrieval framework can be extended to support retrieval-augmented generation (RAG) by combining semantic retrieval with large language models to generate summaries, reports, and answers grounded in enterprise documents. This enables the development of enterprise AI assistants for knowledge management, customer support, and internal decision-making.

These future directions testify to the possibility that the suggested framework can evolve into an overall platform for understanding intelligent documents that has the opportunity to cover a wide range of factors related to enterprise information retrieval and decision support.

REFERENCES

- [1] N. Craswell, B. Mitra, E. Yilmaz, D. Campos, and E. M. Voorhees, "Overview of the TREC 2019 Deep Learning Track," in *Proceedings of the Twenty-Eighth Text REtrieval Conference (TREC)*, 2019.
- [2] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient Estimation of Word Representations in Vector Space," *arXiv preprint arXiv:1301.3781*, 2013.
- [3] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- [4] G. Salton and M. J. McGill, *Introduction to Modern Information Retrieval*. New York, NY, USA: McGraw-Hill, 1983.
- [5] F. Radlinski and N. Craswell, "A Theoretical Framework for Conversational Search," in *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2017.
- [6] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense Passage Retrieval for Open-Domain Question Answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- [7] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey Wolf Optimizer," *Advances in Engineering Software*, vol. 69, pp. 46–61, 2014.
- [8] L. R. Rabiner, "A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
- [9] K. Rathinasamy et al., "EnterpriseEM: Fine-tuned Embeddings for Enterprise Semantic Search," *arXiv preprint arXiv:2406.00010*, 2024.
- [10] P. P. Mishra, K. P. Yeole, R. Keshavamurthy, M. B. Surana, and F. Sarayloo, "A Systematic Framework for Enterprise Knowledge Retrieval: Leveraging LLM-Generated Metadata to Enhance RAG Systems," *arXiv preprint arXiv:2512.05411*, 2025.
- [11] N. Rao, J. Srivastava, P. K. Sharma, and H. Shrivastava, "Scalable and Explainable Enterprise Knowledge Discovery Using Graph-Centric Hybrid Retrieval," *arXiv preprint arXiv:2510.10942*, 2025.
- [12] S. S. Monir, I. Lau, S. Yang, and D. Zhao, "VectorSearch: Enhancing Document Retrieval with Semantic Embeddings and Optimized Search," *arXiv preprint arXiv:2409.17383*, 2024.
- [13] Z. Qiu, Q. Su, J. Yu, and S. Si, "Efficient Document Retrieval by End-to-End Refining and Quantizing BERT Embedding with Contrastive Product Quantization," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- [14] M. L. Sidi and S. Gunal, "A Purely Entity-Based Semantic Search Approach for Document Retrieval," *Applied Sciences*, vol. 13, no. 18, p. 10285, 2023.
- [15] S. V. Mahadevkar et al., "Exploring AI-driven Approaches for Unstructured Document Analysis and Future Horizons," *Journal of Big Data*, 2024, doi:10.1186/s40537-024-00948-z.